



A graph-based LSTM model for PM2.5 forecasting

Xi Gao, Weide Li^{*}

School of Mathematics and Statistics, Center for Data Science, Laboratory of Applied Mathematics and Complex System, Lanzhou University, Lanzhou, 730000, China

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ABSTRACT

Accuracy prediction of air quality is of crucial importance for people to take precautions and improve environmental conditions. By introducing adjacency matrix in Long Short-Term Memory (LSTM) cell, we propose in this research a Graph-based Long Short-Term Memory (GLSTM) model to predict PM2.5 concentration in Gansu Province of Northwest China. We regard all air quality monitoring stations as a graph, and construct a parameterized adjacency matrix on the basis of the adjacency matrix of the graph. Through the combination of parameterized adjacency matrix and LSTM, we introduce spatiotemporal information to achieve PM2.5 prediction. The advantage of GLSTM is that it can realize synchronous operation of all stations, making it unnecessary to train different model for each monitoring station to obtain the overall PM2.5 variation of a certain area. The parameterized adjacency matrix also enhances the interpretability of the model. By visualizing the parameterized adjacency matrix obtained from the end-to-end PM2.5 prediction task in training, the importance of introducing spatial information, i.e. the distribution importance of surrounding stations to a specific station is clearly demonstrated. We compared our model with several newly reported methods, and found that it achieved the best results on PM2.5 prediction tasks at almost all stations, which proved the effectiveness of the GLSTM model.

1. Introduction

Negligence of environmental protection when focusing on economic development leads to air pollution. Large accumulation of air pollutants such as PM2.5 (an indicator of pollutants degree, measuring the fine particulate matter with a diameter of 2.5 μm or less) increases the incidence of cardiovascular and respiratory diseases (Weber et al., 2016; Zhang et al., 2013). Therefore, accurate prediction of PM2.5 can allow the public to take measures in advance, which is of great significance for protecting people's health. Compared with the eastern coastal areas of China, the economically underdeveloped northwestern China with heavy industrial enterprises, land desertification and less vegetation, the smog phenomenon is more obvious and lack of investigation, which needs more research and prediction. This paper takes pollution and PM2.5 of Gansu Province in the northwest region as the research object, and builds a model to predict its PM2.5 concentration.

Current research methods on air pollutant prediction can be categorized into mechanism models and statistical models. Mechanism models mainly study transmission process of air pollutants through meteorological and chemical factors, community multi-scale air quality

model (CMAQ) (Byun and Schere 2006; Yamaji et al., 2008), and European scale Eulerian chemistry transport model (CHIMER-E-continental) (Schmidt et al., 2001) are two of them. Statistical models mainly include time series models and machine learning models. The ARIMA model (Song et al., 2015; Reikard 2012) is the most commonly used method as one in time series models, but it is difficult to introduce other meaningful influence factors for prediction. Machine learning models, which are widely used currently in various fields, and having strong nonlinear fitting capabilities, also have excellent performance in PM2.5 prediction. Zhou et al. (2019) proposed a framework MM-SVM on the basis of SVM, which can build a model for all stations that need to be predicted, avoiding building specific models for each station. As an important field in machine learning, ensemble learning usually combines the advantages of multiple models to complete tasks. An integrated model composed of LASSO, Adaboost, Xgboost and other single models to predict the average daily PM2.5 can get better results than a single model (Zhai and Chen 2018).

Deep learning has been used on PM2.5 prediction recently. As a type of machine learning method, Artificial Neural Network (ANN) has achieved outstanding performance in air quality prediction. Agarwal

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^{*} Corresponding author.

E-mail address: weideli@lzu.edu.cn (W. Li).

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et al. (2020) introduced RTC into ANN and achieved high accuracy in predicting a variety of air pollutants. Convolutional Neural Networks (CNN) and Long Short-term Memory (LSTM) networks being successfully applied to computer vision (Sindagi and Patel 2020; Se et al., 2019) and natural language processing (Guo et al., 2019; Rong 2014) are also gradually applied to PM_{2.5} prediction. As a variant of Recurrent Neural Network (RNN), LSTM can solve problems of RNN—short-term memory and gradient disappearance or explosion, and can better extract time related information in the data. Recently, many studies have constructed new models based on LSTM and achieved excellent results in PM_{2.5} prediction. Chang et al. (2020) aggregate three LSTM models to build an ALSTM model. Seng et al. (2021) proposed a multi-output and multi-index supervised learning model (MMSL). Pak et al. (2020) used Mutual Information (MI) for feature extraction and built a hybrid model of CNN and LSTM to predict daily PM_{2.5}. Model comparison proves that improvement on the basis of the LSTM model to construct a model suitable for the prediction task can achieve better results than the traditional machine learning model. Attention mechanism also becomes a commonly used method in natural language processing (Vaswani et al., 2017). Using attention mechanism to combine target monitoring station with surrounding stations can effectively introduce spatiotemporal information to air pollutants prediction on LSTM model (Shi et al., 2020).

Graph Neural Network (GNN) tunes into an emerging hot field of machine learning. It is mainly used to process graph data, and it has widespread applications in text classification, relation extraction and image classification. GNN is mainly divided into homogeneous graph processing model and heterogeneous graph processing model. Graph Convolutional Neural Network (GCN) (Kipf and Welling 2016) is one of the GNN models which can deal with homogeneous graph, that is, the graph node and edge types are identical. GCN has achieved brilliant performance in paper classification task. Graph Attention Network (GAT) (Veličković et al., 2017) introduces attention mechanism into Graph Neural Network, and uses attention mechanism to aggregate nodes to get better results. GraphSage (Ahmed et al., 2017) used sampling and aggregation methods to classify graph nodes, and contributed to the first commercial use of Graph Neural Networks (Ying et al., 2018). Compared with the aforementioned three models for processing homogeneous graphs, Heterogeneous Graph Neural Network (Zhang et al., 2019) and Heterogeneous Graph Attention Network (HAT) (Wang et al., 2019) are mainly used to process heterogeneous graph data, that is, when types of nodes or edges of the graph are more than just one kind. The combination of GNN and LSTM can process spatiotemporal information in the data more reasonably, and this method is widely used in traffic flow prediction (Yu et al., 2017). In the past, in order to introduce spatial information in air quality prediction, the authors usually concatenated the features from different air quality monitoring stations (Pak et al., 2020). This is not reasonable enough since different stations have different topological structure in space, and simple concatenation may ignore a lot of spatial information. Treating air quality monitoring stations as nodes on the graph and then processing them with graph neural networks can effectively introduce spatial information between different stations in the PM_{2.5} prediction task (Qi et al., 2019).

The Graph-LSTM type model introduces surrounding node information on the basis of LSTM, and is currently used for sentiment classification and semantic relatedness prediction tasks (Li et al., 2015; Tai et al., 2015). Based on some improvements of Tree-LSTM (Tai et al., 2015), Graph-based Long Short-Term Memory (GLSTM) model is proposed in this paper, and is applied to PM_{2.5} prediction. Similar to MM-SVM (Zhou et al., 2019), we propose a general framework that can directly predict PM_{2.5} at all monitoring stations without building a separate model for each station. By combining the advantages of graph neural network processing graph data and LSTM processing time data, we can well introduce spatial information into PM_{2.5} prediction.

2. Data and methods

2.1. Available data

The data in this article mainly comes from China Environmental Monitoring Center, and the hourly air quality data of Gansu Province from November 1, 2019 to October 31, 2020 are selected as the research object of this article. Gansu Province has a long and narrow terrain, located in the upper reaches of the Yellow River in northwestern China. The specific geographical location is shown in Fig. 1 (a). Gansu is an economically underdeveloped province in China, where air pollution is relatively serious due to the influence of heavy industry and the western Taklimakan Desert. This paper selects a total of 32 air quality monitoring stations in Gansu Province for research, and introduces the data of the meteorological station closest to each air quality monitoring station in a straight line as its meteorological features (Shi et al., 2020; Qi et al., 2019). The location information of the stations is shown in Fig. 1 (b). We select 12 variables from the air quality data and meteorological data as the input features of PM_{2.5} prediction. The specific information of the variables is shown in Table 1.

2.2. Methods

2.2.1. Graph construction

We regard the air quality stations in the study area as a graph and construct the adjacency matrix of them. If there is an adjacency between the two air quality stations, then we believe that the data from the two monitoring stations have a strong spatial correlation. The graph composed of air quality stations can be represented by a graph $G = (V, E)$. V represents the set of all nodes in the graph, that is, the set of air quality stations, and E represents the edge in the graph, that is the set of adjacency relations between the stations. Assuming that the graph contains n nodes $v_i, i = 1, \dots, n$, if the edge between two nodes $e_{ij} = (v_i, v_j) \in E$, we consider that there is an adjacency relationship between the two nodes. As shown in formula (1), the adjacency matrix $A \in R^{n \times n}$ means that when v_i and v_j have adjacency relationship, then $A(i, j) = 1$, otherwise it is 0. Since the adjacency matrix does not consider the connection between the node and itself, $A(i, i) = 0$.

$$A(i, j) = \begin{cases} 1 & i \neq j \text{ and } (v_i, v_j) \in E \\ 0 & \text{else} \end{cases} \quad (1)$$

formula (1) constructs a non-weighted adjacency matrix, but in various prediction tasks, the weighted adjacency matrix is usually used to aggregate spatial information.

There are many ways to construct adjacency matrix at present. Lin et al. (2018) used Euclidean distance to calculate the similarity between the “geo-context” of different monitoring stations to construct the adjacency matrix, which is obtained by using random forest to rank the importance of geographic features. Qi et al. (2019) believes that if the distance between two stations is less than a certain constant k , then an adjacency relationship is considered to be in existence and the inverse of the distance is regarded as the element value of the adjacency matrix. Wang et al. (2020) think that there is an adjacency between two cities within a certain distance which requires the mountains between the two cities to be below a certain height, for when the mountains reach a certain height, it may hinder the spread of PM_{2.5}. And they use Heaviside step function to construct the adjacency matrix. However, due to the subjective construction of these graphs, researchers need to obtain the appropriate adjacency matrix through their own prior knowledge or adjustment of hyperparameters. After experiments, introducing such adjacency matrix into the model, the performance is no better than that of other models which are constructed separately by stations. Therefore, we hope that the adjacency relationship between stations can be optimized by end-to-end tasks. We set W_{adj} as a parameter matrix into the model, which means that we believe that there are edges between all

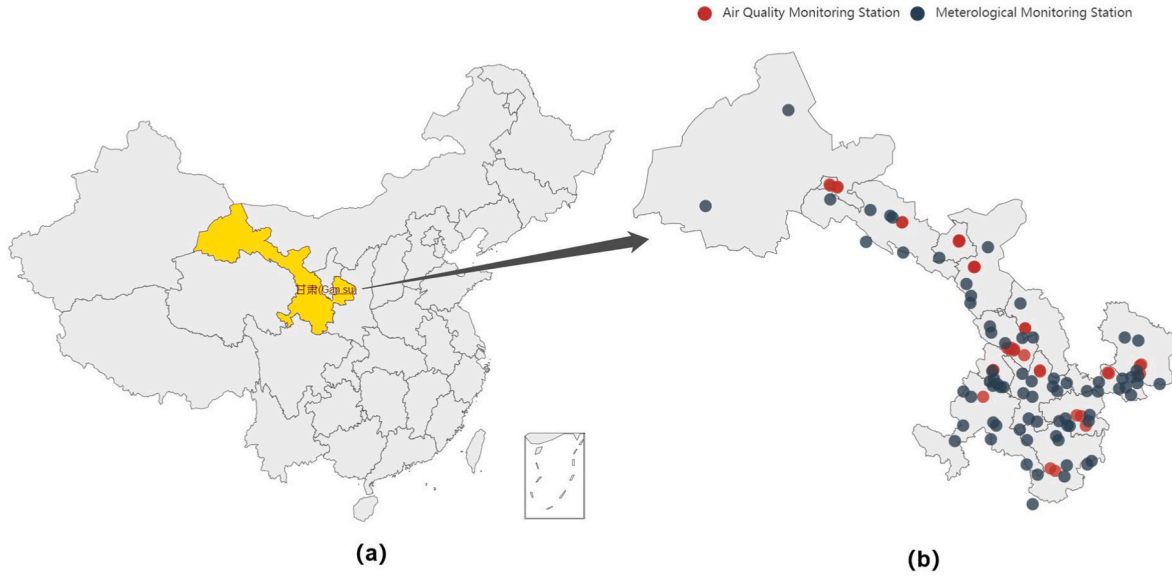


Fig. 1. (a) The location of Gansu Province, China. (b) The locations of all air quality and meteorological monitoring stations in Gansu Province. The red dots are air quality monitoring stations and the blue dots are meteorological monitoring stations.

Table 1
The properties of the variables.

Variable	Unit	Range	Mean	Std
CO	mg/m ³	[0.1, 8.8]	0.650	0.489
NO ₂	μg/m ³	[1, 270]	25.007	20.111
O ₃	μg/m ³	[1, 507]	66.622	36.284
PM ₁₀	μg/m ³	[1, 9985]	67.323	36.284
PM _{2.5}	μg/m ³	[1, 1276]	28.063	29.477
SO ₂	μg/m ³	[1, 869]	12.645	19.533
Temperature	°C	[− 21.1, 36.4]	9.446	10.799
Pressure	hPa	[35.53, 995.20]	844.471	35.527
Relative humidity	%	[0, 100]	55.366	26.589
Wind direction	°	[0, 360]	185.908	106.560
Wind speed	m/s	[0, 21]	2.451	1.583
Precipitation	mm	[0.0, 40.3]	0.044	0.418

nodes, and the weights of the edges are related to the correlation between the nodes. Usually the parameter adjacency matrix W_{adj} obtained by model training is an asymmetric matrix, that is to say, the graph G composed of air quality stations is a directed graph ($e_{ij} \neq e_{ji}$). This means that when spatial information is introduced, the importance of monitoring station v_i to monitoring station v_j is not the same as the importance of monitoring station v_j to monitoring station v_i .

2.2.2. GLSTM layer

We use $X_t \in R^{c \times n}$ to represent the c air quality and meteorological variables of n stations at time t , and $Y_t \in R^n$ to represent the PM_{2.5} concentration of n stations at time t . In order to introduce time information, we consider using $[X_{t-1}, X_{t-1+1}, \dots, X_{t-1}]$ to predict Y_t . Unlike Tree-LSTM (Tai et al., 2015), we use W_{adj} to introduce the adjacency relationship between different stations, so we only assign a forget gate to each station instead of adding forget gates to all edges, which is also helpful to reduce the amount of parameters. Then formulas (2)–(7) are obtained. In this way, we introduced spatiotemporal information in the PM_{2.5} forecasting task and realized the synchronization operation of all stations.

$$i_v^t = \sigma \left(W^i x_v^t + U^i \sum_{k=1}^n w_{k,v} h_k^{t-1} + b^i \right) \quad (2)$$

$$f_v^t = \sigma \left(W^f x_v^t + U^f \sum_{k=1}^n w_{k,v} h_k^{t-1} + b^f \right) \quad (3)$$

$$o_v^t = \sigma \left(W^o x_v^t + U^o \sum_{k=1}^n w_{k,v} h_k^{t-1} + b^o \right) \quad (4)$$

$$u_v^t = \tanh \left(W^u x_v^t + U^u \sum_{k=1}^n w_{k,v} h_k^{t-1} + b^u \right) \quad (5)$$

$$c_v^t = i_v^t \odot u_v^t + \sum_{k=1}^n w_{k,v} f_v^t \odot c_k^{t-1} \quad (6)$$

$$h_v^t = o_v^t \odot \tanh(c_v^t) \quad (7)$$

Same as LSTM structure, GLSTM also consists of input gate, forget gate and output gate. It can be seen that when the number of stations is reduced to one, the GLSTM structure is approximately degraded to LSTM. Different from LSTM, in the process of information flowing over time, the information of neighbor stations will be introduced according to the weight matrix W_{adj} . The edge from station k to station v is represented by $w_{k,v} = W_{adj}(k, v)$. The input gates $i_v^t \in R^f$ and $u_v^t \in R^f$ are used to control whether to introduce the input vector $x_v^t \in R^c$ and the hidden state $h_k^{t-1} \in R^f, k = 1, \dots, n$ of all stations at time $t - 1$. The forget gate $f_v^t \in R^f$ is used to control whether to introduce the memory information of $t - 1$ itself and other stations. The output gate $o_v^t \in R^f$ outputs the hidden vector h_v^t by controlling the memory state $c_v^t \in R^f$ of the station v at time t . In formulas (2)–(5), $W^i, W^f, W^o, W^u \in R^{f \times c}$ are the parameter matrix for controlling the input information, $U^i, U^f, U^o, U^u \in R^{f \times f}$ are the parameter matrix for controlling the hidden state, $b^i, b^f, b^o, b^u \in R^f$ are bias vectors, $\odot$ is Hadamard product. A GLSTM cell with three stations is shown in Fig. 2, where v is the target station, v_1 and v_2 are the introduced neighbor stations, $h_v^{t-1}, h_{v_1}^{t-1}, h_{v_2}^{t-1}$ and $c_v^{t-1}, c_{v_1}^{t-1}, c_{v_2}^{t-1}$ are the corresponding hidden states and memory states at $t - 1$. x_v^t and h_v^t are the input vector and hidden state of station v at time t . The hidden state and memory state of the three stations that are multiplied by the adjacency matrix at time $t - 1$ and the information of the station v at time t are integrated in a GLSTM cell according to formulas (2)–(7).

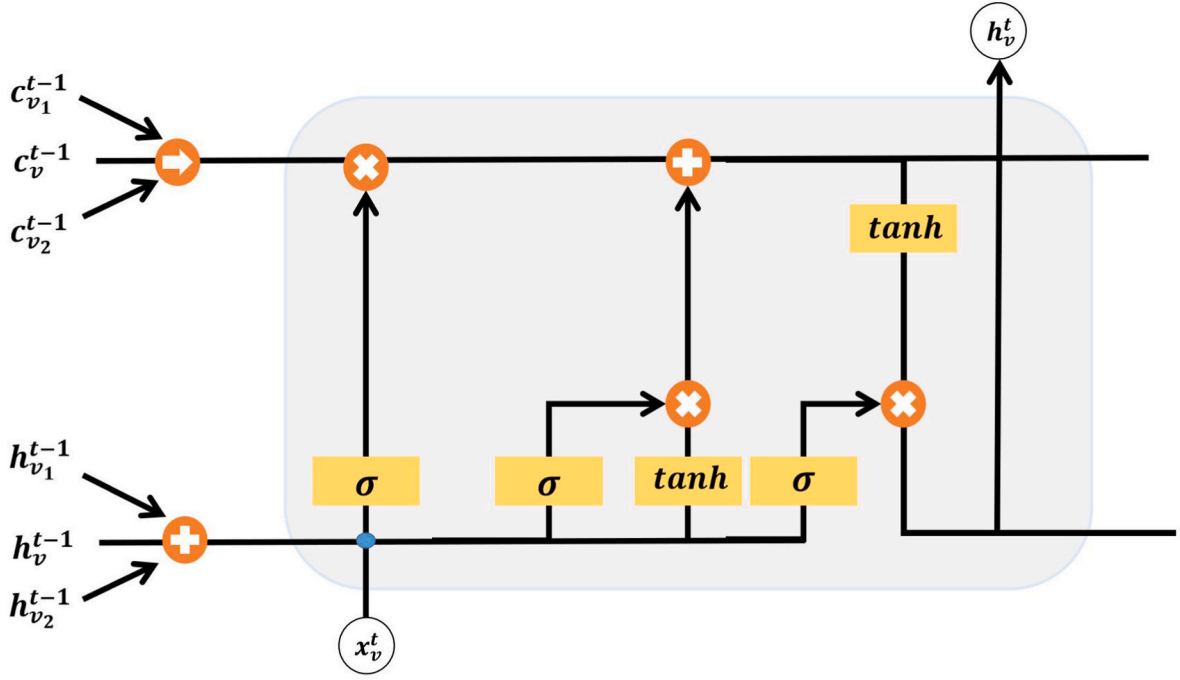


Fig. 2. A structure of GLSTM cell on station v.

The process of GLSTM synchronization operation for n stations is shown in formulas (8)-(13).

$$I_t = \sigma(W^i X_t + U^i H_{t-1} W_{adj} + B^i) \quad (8)$$

$$F_t = \sigma(W^f X_t + U^f H_{t-1} + B^f) \quad (9)$$

$$O_t = \sigma(W^o X_t + U^o H_{t-1} W_{adj} + B^o) \quad (10)$$

$$U_t = \tanh(W^u X_t + U^u H_{t-1} W_{adj} + B^u) \quad (11)$$

$$C_t = I_t \odot U_t + F_t \odot C_{t-1} W_{adj} \quad (12)$$

$$H_t = O_t \odot \tanh(C_t) \quad (13)$$

$I_t \in R^{f \times n}$ and $U_t \in R^{f \times n}$ are the control variable and output matrix of the input gates of n stations at time t , $F_t \in R^{f \times n}$ and $O_t \in R^{f \times n}$ are the control variables of the forget gate and output gate at time t , $C_t \in R^{f \times n}$ and $H_t \in R^{f \times n}$ are the memory states and hidden states of n stations, $B^i, B^f, B^o, B^u \in$

$R^{f \times n}$ are the matrices obtained by copying the bias vectors b_i, b_f, b_o, b_u with n times. Through formulas (8)-(13), we can see that all stations in a GLSTM layer share same parameters $W^i, W^f, W^o, W^u \in R^{f \times c}$, $U^i, U^f, U^o, U^u \in R^{f \times f}$ and $B^i, B^f, B^o, B^u \in R^{f \times n}$. And each GLSTM layer has only one W_{adj} to aggregate neighbor information. A GLSTM layer containing three stations is shown in Fig. 3. There are edges with weights to be optimized between stations v_1, v_2 and v_3 . If v_3 is not important to v_1 , then $w_{3,1}$ will be approximately 0 as the model is trained. If v_1 and v_2 are important to v_3 , $w_{1,3}$ and $w_{2,3}$ will gradually become larger as the training process.

2.2.3. Model construction

As shown in Fig. 4, we built a model with a GLSTM layer and a fully connected layer. The number of hidden layer nodes of the GLSTM layer is set to 15, and the initial hidden state H_0 and C_0 of the GLSTM layer are set to zero matrix; the number of nodes of the fully connected layer is set to 1. Since the forward propagation mechanism of the GLSTM layer and the LSTM layer is more complex, the deep models of the two have difficulty in training and poor prediction performance compared to shallow networks. Therefore, we set the number of GLSTM layers and fully

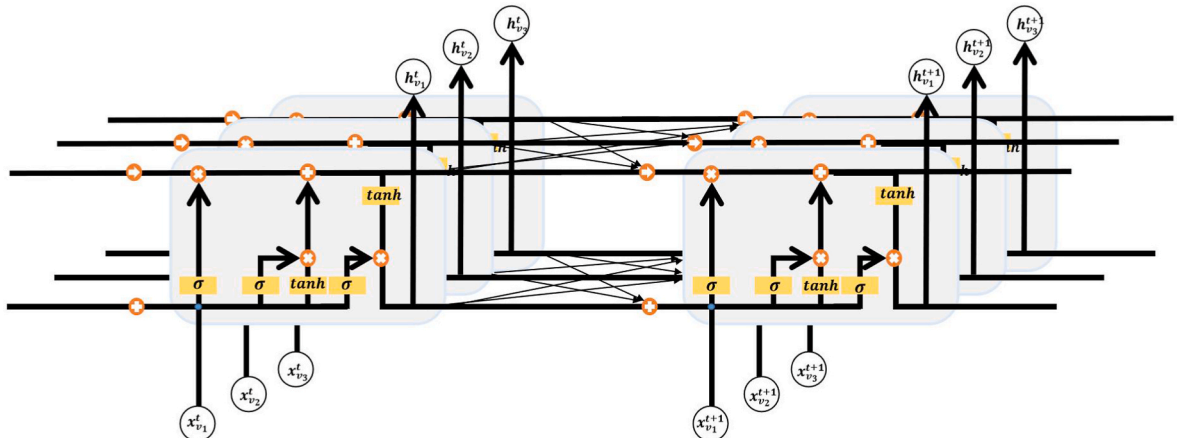


Fig. 3. GLSTM layer structure containing three stations.

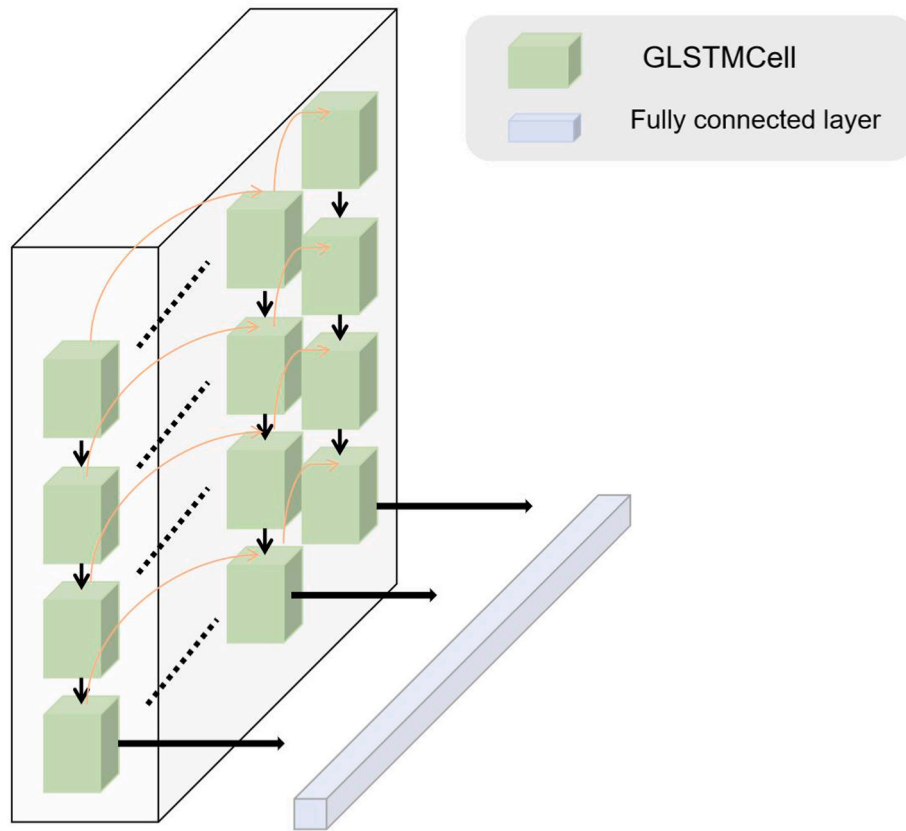


Fig. 4. GLSTM model structure.

connected layers to 1. We set the time length l of the feature variable to 4, that is, use $[X_{t-4}, X_{t-3}, X_{t-2}, X_{t-1}]$ to predict Y_t . Before inputting these data into the model, we use formula (16) to normalize it for each station. The block on the left in Fig. 4 is GLSTM layer. The vertical direction of the layer is the direction of time flow, the horizontal direction represents different stations, and the orange arrows represent the adjacency between the stations. For the GLSTM layer, we only take the output of the last moment as the input of the fully connected layer. As shown in formulas (14) and (15), the input of the entire model is in the format $[N, n, l, c]$, where N is the number of samples, n is the number of stations, l is the length of the data period, c is the feature dimension; the output format is $[N, n]$. The detailed information of the whole model is shown in Table 2.

$$\text{Input} = \begin{bmatrix} X_0 & X_1 & \cdots & X_{l-1} \\ X_1 & X_2 & \cdots & X_l \\ \vdots & \vdots & \ddots & \vdots \\ X_{N-1} & X_N & \cdots & X_{N+l-2} \end{bmatrix} \quad (14)$$

Table 2
Model information.

	Parameter	Description	Value
Input	N	Number of samples	8780
	n	Number of stations	32
	l	Length of time	4
	c	Feature dimension	12
Output	N	Number of samples	8780
	n	Number of stations	32
Model	GLSTM_1	Number of parameters of the first GLSTM layer	2704
	Dense_1	Number of parameters of the first fully connected layer	15

$$\text{output} = \begin{bmatrix} Y_l \\ Y_{l+1} \\ \vdots \\ Y_{l+N-1} \end{bmatrix} \quad (15)$$

$$x_{\text{norm}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \quad (16)$$

2.2.4. Experimental settings

We split the data set according to the time dimension, with the first 60 % as the training set, then 20 % as the validation set, and the last 20 % as the test set. The maximum training epochs of the model is set to 1000; the learning rate is set to 0.05; the weight decay rate is set to $5e-4$; the loss function is set to Mean Absolute Error (MAE); and the optimizer is set to Adaptive moment estimation (Adam). We monitor the loss of the validation set and stop the training process early. When the training is greater than 200 epochs, we start to use the early stop mechanism to save the model parameters that perform best on the validation set during the model training process. The patience of the early stopping mechanism is set to 100, that is, when the loss of the validation set is greater than the optimal loss 100 times in a row, we stop the training process.

Table 3
Details of the experimental settings.

Parameter	Value
Training set	60 % (2019.11.01 00:00–2020.06.07 10:00)
Validation set	20 % (2020.06.07 11:00–2020.08.19 14:00)
Test set	20 % (2020.08.19 15:00–2020.10.31 23:00)
Maximum epochs of training	1000
Learning rate	0.05
Weight decay rate	0.0005
Patience of early stopping mechanism	100
Loss function	Mean Absolute Error

The details of the experimental setup are shown in Table 3. We choose Root Mean Square Error (RMSE) (17), Mean Absolute Error (MAE) (18), Mean Absolute Percentage Error (MAPE) (19) and IA (20) as metrics to evaluate the performance of GLSTM model on each station to compare with other models.

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (17)$$

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (18)$$

$$MAPE = \frac{1}{N} \sum_{i=1}^N \frac{|y_i - \hat{y}_i|}{y_i} \quad (19)$$

$$IA = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (|\hat{y}_i - \bar{y}| + |y_i - \bar{y}|)^2} \quad (20)$$

where y_i , $\hat{y}_i$ and N represent the observed values, the predicted value and the number of evaluation samples at each station, respectively. $\bar{y}$ is average value of N observed sample data. RMSE, MAE and MAPE are commonly used evaluation metrics in time series forecasting. The lower their value, the better the model performance. IA is a nondimensional and bounded measure of the model prediction error degree with values closer to 1 indicating a better match (Qi et al., 2019).

3. Results and discussion

We first use the GLSTM model and the compared models to predict PM2.5 in one step, that is, predict the PM2.5 concentration in the next hour. The followings are the comparisons:

- **eXtreme Gradient Boosting (XGBoost)**: XGBoost is an ensemble learning model, which is widely used in various machine learning competitions, and the model performance is usually better than

random forest (Hill et al., 2020). Due to the requirements of the model input format, we model and predict the data of each air quality monitoring station separately, and flatten the data format of every sample into one dimension.

- **LSTM**: LSTM is a widely used model in time series forecasting (Shi et al., 2020), and it is also the basis for our model. We also model and forecast the data of each air quality monitoring station separately. In keeping with the GLSTM model, the LSTM model is also composed of an LSTM layer and a fully connected layer.
- **GC-LSTM**: GC-LSTM is composed of graph convolutional neural network and LSTM, and predicts the spatio-temporal sequence of PM2.5 (Qi et al., 2019). The adjacency matrix of the model is constructed as shown in formula (21). When the straight-line distance d_{ij} between air quality monitoring stations i and j is less than R , we consider that there is an adjacency relationship. After experiments, we set R to 400 km, and the adjacency relationship between the stations is shown in Fig. 5.
- **GC-DCRNN**: Lin et al. (2018) replaced the matrix multiplication in the gated linear unit (GRU) with diffusion convolution and proposed the GC-DCRNN model. The adjacency matrix of the model is constructed based on the geographical features around the air quality monitoring station. In this article, we use the same adjacency matrix construction method as GC-LSTM.
- **I-GLSTM**: We set the adjacency matrix of the GLSTM model as the identity matrix to obtain I-GLSTM, which means that the model only considers its own information without introducing information from neighboring stations.
- **D-GLSTM**: In order to reflect the benefits of our model in introducing parameterized adjacency matrix, we also use formula (21) to construct the adjacency matrix to obtain D-GLSTM, which is also used as one of the comparison models.

$$A_{ij} = \begin{cases} \frac{1}{d_{ij}}, & i \neq j \wedge d_{ij} < R \\ 0, & \text{otherwise} \end{cases} \quad (21)$$

We use the segmented test set in Section 2.2.4 to verify the performance of the model. In the main text, we only show the prediction results of 6 air quality monitoring stations, and the complete results are in the appendix. The comparison between the predicted value of each



Fig. 5. Visualization of the adjacency matrix used by GC-LSTM, GC-DCRNN and D-GLSTM. The edges represent the adjacency between different monitoring stations, and the arrows represent the direction of the edges.

model and the true value is shown in Fig. 6. Because there are many comparison models, it is not easy to show the differences between the models with one diagram, but we can still see that most models can predict the change trend of PM_{2.5} well. In order to better observe the performance of our model, the prediction results of it and the real value at all air quality monitoring stations are shown in the appendix.

Since the prediction results of deep learning models usually fluctuate, we conducted 10 experiments and showed the mean and standard deviation (std) of the four metrics as formulas (17)–(19) in Table 4. The performance of XGBoost when the parameters and data are fixed and does not change, that is, the std is 0, so we did not show its std in the table. The bolded numbers in Table 4 are the best results of the experiment. It can be seen that the GLSTM model can get the best results in the PM_{2.5} predictions of most air quality monitoring stations. XGBoost, as a traditional machine learning model, performs worse than LSTM-type deep learning models. Due to the lack of introduced spatial information, LSTM also did not get the best results. Most of the performances of I-GLSTM, D-GLSTM and GLSTM are better than GC-LSTM and GC-DCRNN, which proves the superiority of our model. GLSTM constructs a parameter-sharing LSTM for each station and uses the adjacency matrix for spatial information aggregation. A GLSTM layer can extract spatiotemporal sequence information without combining multiple models. I-GLSTM only considers the information of the monitoring station itself and ignores the spatial information, so its model performance does not exceed D-GLSTM and GLSTM. D-GLSTM only considers the distance relationship between different air quality monitoring stations when constructing the adjacency matrix, ignoring the similarity of geography and economic environment between some stations, so the performance is worse than GLSTM. GLSTM uses a parameterized adjacency matrix, which allows the model to be optimized according to end-to-end tasks when introducing spatial information to achieve the best results. Some of the prediction results of GC-LSTM, GC-DCRNN, I-GLSTM and D-GLSTM are not even as good as the LSTM modeled separately by each monitoring station. This is largely due to the fact that the identity matrix (I-GLSTM) and the adjacency matrix constructed based on formula (21) (GC-LSTM, GC-DCRNN, D-GLSTM) cannot meet the prediction task. The graph-based spatiotemporal sequence prediction model uses the adjacency matrix to realize the aggregation of spatial

information. The performance of this type of model largely depends on the construction of the adjacency matrix. The adjacency matrix constructed with prior knowledge is not only difficult to consider comprehensively, but also easy to introduce wrong information, which will reduce the performance of the model. In contrast, the parameterized adjacency matrix is more reasonable and convenient. Although the performance of the GLSTM model in the appendix on all air quality monitoring stations is not as good as that of the 6 stations shown in Table 4, it still obtains the best results in most predictions.

We can see that when there are differences in model performance, the trends of the four metrics are not completely consistent. Taking the 1945A monitoring station as an example, the RMSE, MAE and IA of the GLSTM model are better than other models, but the opposite occurred for MAPE. RMSE is usually affected by extreme values, so the performance is not completely consistent with MAE. For MAE and MAPE, if the model performs slightly worse on samples with smaller values, the MAE of the model is not very high on this part of the sample, but MAPE may also be higher than other models. After experiments, the IA will be higher when the distribution of the predicted value is close to the true value, and the greater the difference between the predicted value and the true value is, the lower the IA. So it is very meaningful to comprehensively consider the four metrics to evaluate the model. GLSTM achieves the best performance on almost all the four metrics at all stations, which also proves the effectiveness of the model. And after comparison, it is found that the GLSTM model excel itself when predicting high concentration of PM_{2.5}.

Since we introduce the spatial information between different stations through a parametric adjacency matrix W_{adj} that can be optimized, this approach enhances the interpretability of the model. We can look at the parameter values of the adjacency matrix W_{adj} obtained at the end to get the importance of the relationship between different stations to find out whether the spatial information is well introduced. We first use formula (16) to normalize W_{adj} by column, and then multiply it by 100 to obtain the adjacency matrix with a value range of [0, 100]. We select two stations for visualization, and the results are shown in Figs. 7 and 8. As we mentioned in subsection 2.2.1, W_{adj} is not approximately a symmetric matrix, which means that the importance of stations v_i and v_j to

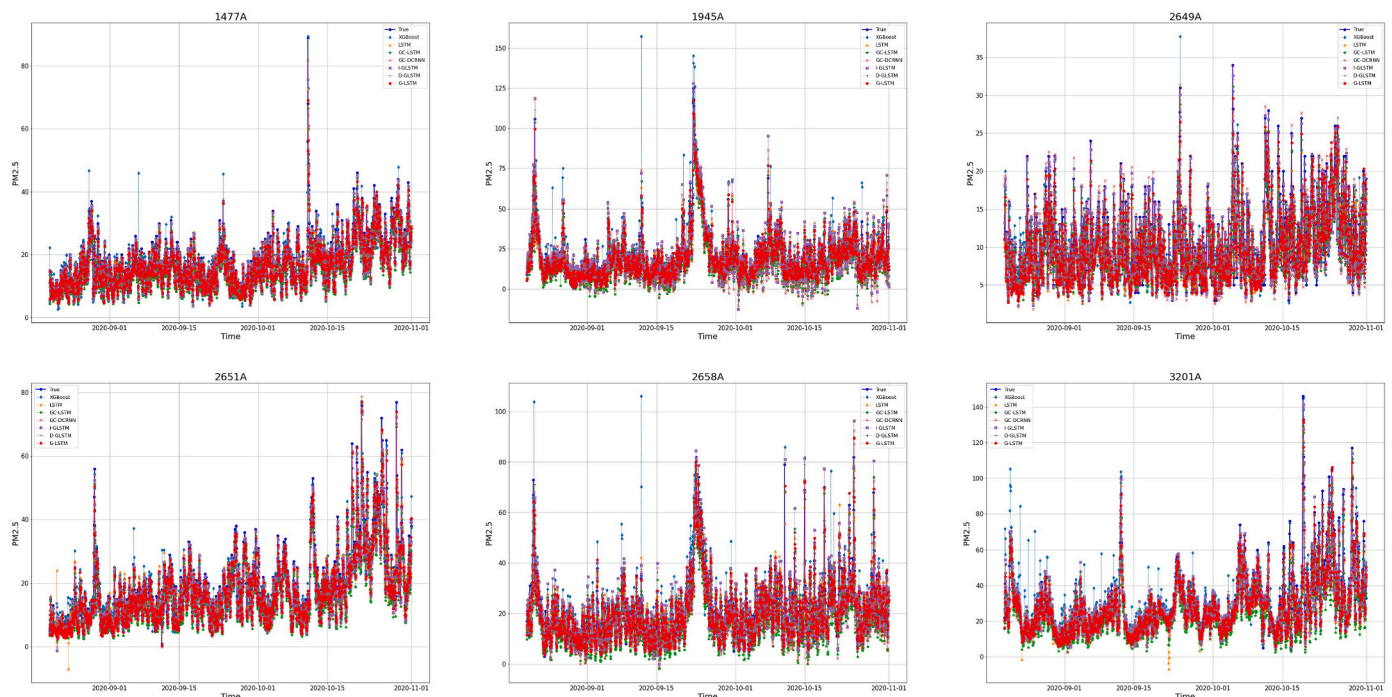
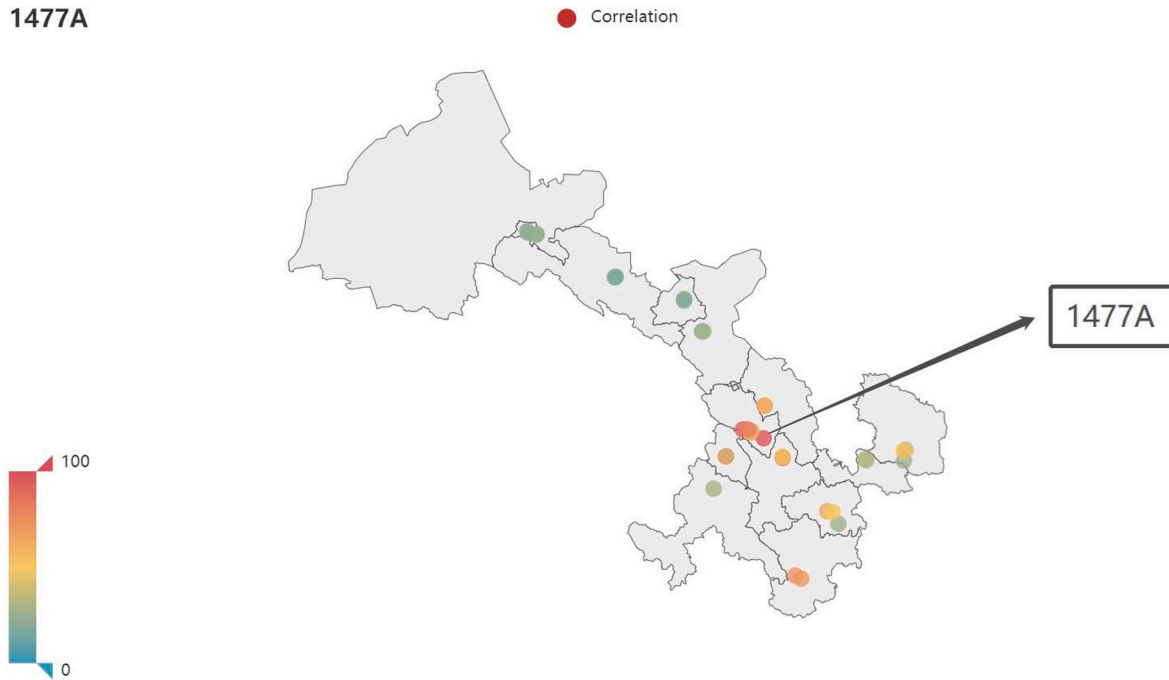


Fig. 6. Comparison of different model predicted values and true value.

Table 4

The predicted performance of different models at 6 air quality monitoring stations, and the bolded values in the table are the best results.

Model	Metric	1478A	1945A	2650A	2652A	2660A	3201A
XGBoost	RMSE	5.968	7.535	3.907	7.828	6.246	7.653
	MAE	4.572	4.489	3.036	4.552	4.314	4.818
	MAPE	0.306	0.275	0.274	0.203	0.455	0.185
	IA	0.899	0.933	0.949	0.923	0.941	0.943
LSTM	RMSE	5.111 (0.026)	7.141 (0.375)	3.493 (0.030)	10.84 (1.665)	5.367 (0.020)	6.866 (0.696)
	MAE	3.700 (0.018)	4.650 (0.242)	2.659 (0.021)	7.275 (1.393)	3.113 (0.031)	4.276 (0.481)
	MAPE	0.198 (0.001)	0.260 (0.014)	0.209 (0.005)	0.283 (0.040)	0.194(0.002)	0.141 (0.015)
	IA	0.929 (0.001)	0.924 (0.013)	0.960 (0.001)	0.790 (0.082)	0.958 (0.000)	0.945 (0.016)
GC-LSTM	RMSE	5.008 (0.057)	7.207 (0.850)	3.555 (0.066)	7.500 (1.239)	5.507 (0.073)	6.448 (0.565)
	MAE	3.621 (0.029)	5.168 (0.876)	2.704 (0.035)	5.347 (1.417)	3.323 (0.116)	4.369 (0.585)
	MAPE	0.196 (0.004)	0.330 (0.073)	0.210 (0.007)	0.261 (0.088)	0.248 (0.039)	0.157 (0.026)
	IA	0.935 (0.002)	0.934 (0.014)	0.959 (0.002)	0.930 (0.021)	0.955 (0.002)	0.954 (0.008)
GC-DCRNN	RMSE	4.978 (0.080)	8.274 (1.001)	3.519 (0.044)	7.720 (0.386)	5.427 (0.117)	6.781 (0.941)
	MAE	3.574 (0.053)	6.049 (1.049)	2.675 (0.023)	5.304 (0.537)	3.226 (0.212)	4.852 (1.073)
	MAPE	0.191(0.002)	0.386 (0.077)	0.209 (0.003)	0.250 (0.034)	0.207 (0.015)	0.180 (0.045)
	IA	0.932 (0.002)	0.921 (0.018)	0.961 (0.001)	0.933 (0.006)	0.958 (0.002)	0.952 (0.013)
I-GLSTM	RMSE	5.142 (0.024)	7.749 (0.385)	3.539 (0.017)	7.220 (0.386)	5.422 (0.016)	5.850 (0.199)
	MAE	3.723 (0.013)	5.390 (0.429)	2.695 (0.009)	4.850 (0.472)	3.097 (0.035)	3.914 (0.212)
	MAPE	0.201 (0.002)	0.343 (0.039)	0.211 (0.004)	0.228 (0.031)	0.207 (0.014)	0.145 (0.011)
	IA	0.933 (0.001)	0.929 (0.007)	0.960 (0.000)	0.937 (0.007)	0.958 (0.000)	0.966 (0.003)
D-GLSTM	RMSE	4.984 (0.076)	6.465 (0.111)	3.469 (0.063)	6.734 (0.274)	5.363 (0.125)	6.044 (0.369)
	MAE	3.617 (0.063)	4.406 (0.156)	2.649 (0.051)	4.279 (0.312)	3.091 (0.103)	3.958 (0.359)
	MAPE	0.198 (0.006)	0.255(0.008)	0.210 (0.005)	0.180 (0.014)	0.203 (0.007)	0.137 (0.013)
	IA	0.936 (0.001)	0.945 (0.003)	0.961 (0.001)	0.941 (0.007)	0.958 (0.002)	0.962 (0.005)
GLSTM	RMSE	4.864(0.017)	6.267(0.124)	3.405(0.020)	6.458(0.148)	5.260(0.039)	5.733(0.163)
	MAE	3.528(0.012)	4.233(0.126)	2.606(0.017)	3.864(0.184)	3.091(0.064)	3.617(0.188)
	MAPE	0.194 (0.002)	0.262 (0.016)	0.209(0.003)	0.161(0.009)	0.216 (0.013)	0.126(0.006)
	IA	0.938(0.001)	0.948(0.002)	0.963(0.000)	0.946(0.003)	0.960(0.001)	0.966(0.002)

**Fig. 7.** Correlation of 1477A with surrounding stations.

each other is not consistent. For example, in Figs. 7 and 8, the importance of stations 1477A and 2665A to each other is not the same. It can also be seen that the parameter adjacency matrix W_{adj} better reflects the spatial information. The air quality monitoring station 2665A in Fig. 8 is located in the Longnan area of Gansu Province, which is the only semi-humid area in Gansu Province, in sharp contrast with the rest of the semi-arid areas in Gansu Province. Therefore, 2665A has the strongest correlation with the nearest air quality station. At the same time, the parameterized adjacency matrix contains more than just distance

information.

Take 1477A in Fig. 7 as an example. It is located in Lanzhou, the most economically developed area of Gansu Province. It has a strong correlation with most air quality monitoring stations in southern Gansu Province, but has a weak correlation with stations in northern Gansu Province. This is mainly due to the Wushaoling Mountains in the central part of Gansu Province. PM2.5 in the northern part of Gansu Province usually hardly affects the southern part of Gansu Province. The parameterized adjacency matrix W_{adj} plays a role similar to the attention

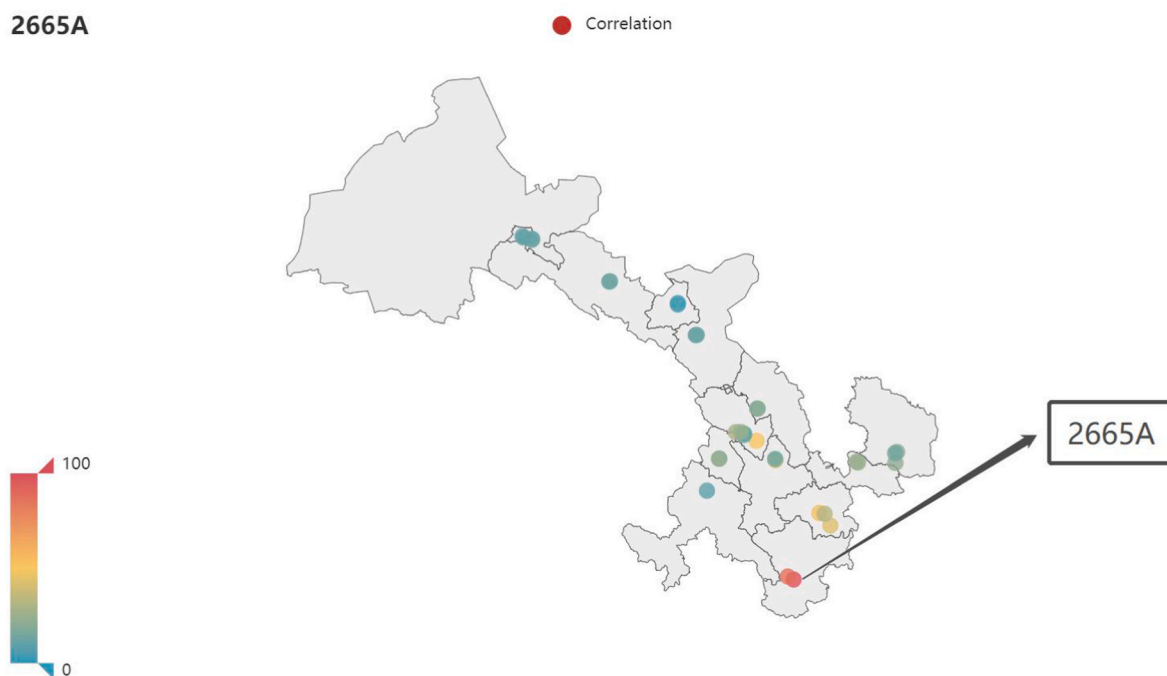


Fig. 8. Correlation of 2665A with surrounding stations.

mechanism of neighbor nodes in this article, which allows the model to pay more attention to the most valuable neighbor information according to the prediction task.

In order to reflect the advantages of the GLSTM model for simultaneous operation of all stations, we randomly select the prediction results of 5 consecutive moments in the test set to compare with the real results, and visualized them in the form of heat maps. The results are shown in Fig. 9. The first five pictures of Fig. 9 are the predictions of the model, and the last five pictures are the real changes of PM2.5. It can be seen that the overall change of PM2.5 in Gansu Province predicted by the GLSTM model is very close to the real situation. Since October 1st is China's National Day, more vehicles emit more exhaust to cause more serious air pollution, we can find that the PM2.5 concentration is generally high, and the model can also have very good prediction at this time.

In order to observe the long-term predictive ability of the model, we select 7 different times to use the model for multi-step forecasting. We still segment the dataset as being described in Table 3, and use formulas (17)–(20) to evaluate the model results. Since displaying the prediction results of all air quality monitoring stations will make the table too large, we choose to average the prediction results, that is, to calculate the average error of all samples at each prediction time step. The results are shown in Table 5. It can be seen that the GLSTM model is better than the comparative model in the short-term and long-term predictions. But when GLSTM is applied to predict PM2.5 concentration for 4 h or longer, its performance on the IA index is not as good as GC-DCRNN. We separately compared the 12h prediction results of GLSTM and GC-DCRNN at air quality monitoring stations 2647A and 2652A (Fig. 10). The result of GC-DCRNN in the long-term prediction fluctuates more drastically, while the prediction result of GLSTM is smoother. This will

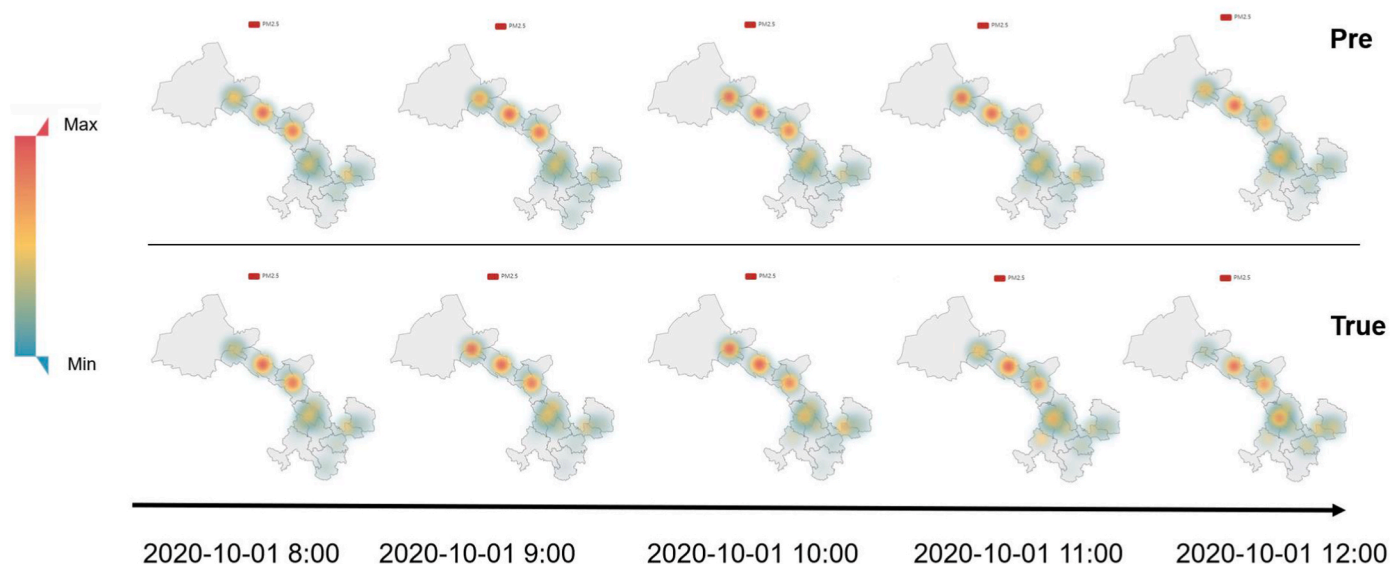


Fig. 9. Heat map of GLSTM model's overall PM2.5 forecast in Gansu Province.

Table 5
Results of multi-step predictions of different models.

Model	Metric	+1 h	+2 h	+4 h	+8 h	+12 h	+24 h	+48 h
XGBoost	RMSE	6.452	9.62	13.246	16.341	17.145	18.291	18.128
	MAE	4.129	6.128	8.459	10.463	11.334	11.802	11.914
	MAPE	0.312	0.48	0.685	0.853	0.956	0.983	1.007
	IA	0.921	0.828	0.671	0.557	0.51	0.486	0.473
LSTM	RMSE	5.418 (0.120)	7.287 (0.076)	9.476 (3.264)	10.761 (3.629)	11.128 (3.676)	10.959 (3.666)	11.185 (3.622)
	MAE	3.545 (0.080)	4.726 (0.055)	6.228 (1.636)	7.151(1.940)	7.433 (2.037)	7.336 (1.979)	7.794 (2.098)
	MAPE	0.220 (0.004)	0.300 (0.003)	0.394 (0.116)	0.454 (0.131)	0.482 (0.140)	0.513 (0.183)	0.572 (0.215)
	IA	0.936 (0.005)	0.880 (0.004)	0.753 (0.088)	0.632 (0.116)	0.590 (0.113)	0.583 (0.131)	0.542 (0.139)
GC-LSTM	RMSE	5.365 (0.161)	7.677 (0.297)	9.962 (3.334)	11.299 (3.673)	11.846 (3.901)	11.813 (4.054)	12.741 (4.504)
	MAE	3.616 (0.195)	5.175 (0.338)	6.565 (1.681)	7.511 (1.924)	7.999 (2.211)	8.031 (2.395)	8.801 (2.893)
	MAPE	0.232 (0.024)	0.332 (0.045)	0.399 (0.094)	0.453(0.124)	0.487 (0.137)	0.510 (0.152)	0.562 (0.183)
	IA	0.943 (0.004)	0.875 (0.010)	0.756 (0.081)	0.650 (0.094)	0.617 (0.092)	0.630 (0.102)	0.551 (0.124)
GC-DCRNN	RMSE	5.787 (0.277)	7.679 (0.018)	9.912 (3.391)	11.256 (3.808)	11.590 (4.033)	11.679 (4.014)	12.429 (4.409)
	MAE	3.945 (0.305)	5.107 (0.012)	6.648 (1.901)	7.559 (2.075)	7.931 (2.404)	7.951 (2.377)	8.559 (2.787)
	MAPE	0.248 (0.016)	0.324 (0.002)	0.426 (0.116)	0.462 (0.124)	0.509 (0.157)	0.505 (0.157)	0.561(0.190)
	IA	0.934 (0.005)	0.879 (0.001)	0.773(0.070)	0.673(0.084)	0.657(0.088)	0.636(0.092)	0.568 (0.110)
I-GLSTM	RMSE	5.309 (0.068)	7.604 (0.057)	10.145 (3.449)	11.563 (3.825)	11.964 (4.027)	11.792 (3.972)	12.564 (4.332)
	MAE	3.514 (0.084)	5.015 (0.045)	6.803 (1.941)	7.763 (2.068)	8.131 (2.326)	7.977 (2.298)	8.642 (2.731)
	MAPE	0.223 (0.011)	0.315 (0.004)	0.426 (0.109)	0.466 (0.116)	0.499 (0.136)	0.501 (0.146)	0.564 (0.190)
	IA	0.946 (0.001)	0.883 (0.001)	0.762 (0.072)	0.657 (0.086)	0.630 (0.089)	0.624 (0.098)	0.551 (0.117)
D-GLSTM	RMSE	5.240 (0.068)	7.095 (2.416)	9.485 (3.393)	10.999 (3.869)	11.239 (3.933)	10.847 (3.826)	11.201 (4.035)
	MAE	3.453 (0.051)	4.669 (1.209)	6.305 (1.807)	7.396 (2.217)	7.617 (2.251)	7.351 (2.140)	7.858 (2.407)
	MAPE	0.212 (0.004)	0.294 (0.065)	0.392(0.093)	0.464 (0.132)	0.481 (0.126)	0.495 (0.131)	0.578 (0.168)
	IA	0.945 (0.001)	0.889 (0.040)	0.755 (0.087)	0.610 (0.123)	0.605 (0.122)	0.614 (0.138)	0.573 (0.156)
GLSTM	RMSE	5.082(0.027)	7.088(0.031)	9.357(3.319)	10.757(3.702)	10.929(3.806)	10.838(3.889)	11.101(3.973)
	MAE	3.331(0.025)	4.666(0.022)	6.224(1.727)	7.245 (2.038)	7.355(2.145)	7.255(2.195)	7.769(2.398)
	MAPE	0.210(0.003)	0.294(0.003)	0.393 (0.090)	0.459 (0.124)	0.476(0.133)	0.493(0.133)	0.572 (0.171)
	IA	0.948(0.001)	0.889(0.001)	0.762 (0.086)	0.639 (0.120)	0.605 (0.134)	0.615 (0.136)	0.575(0.154)

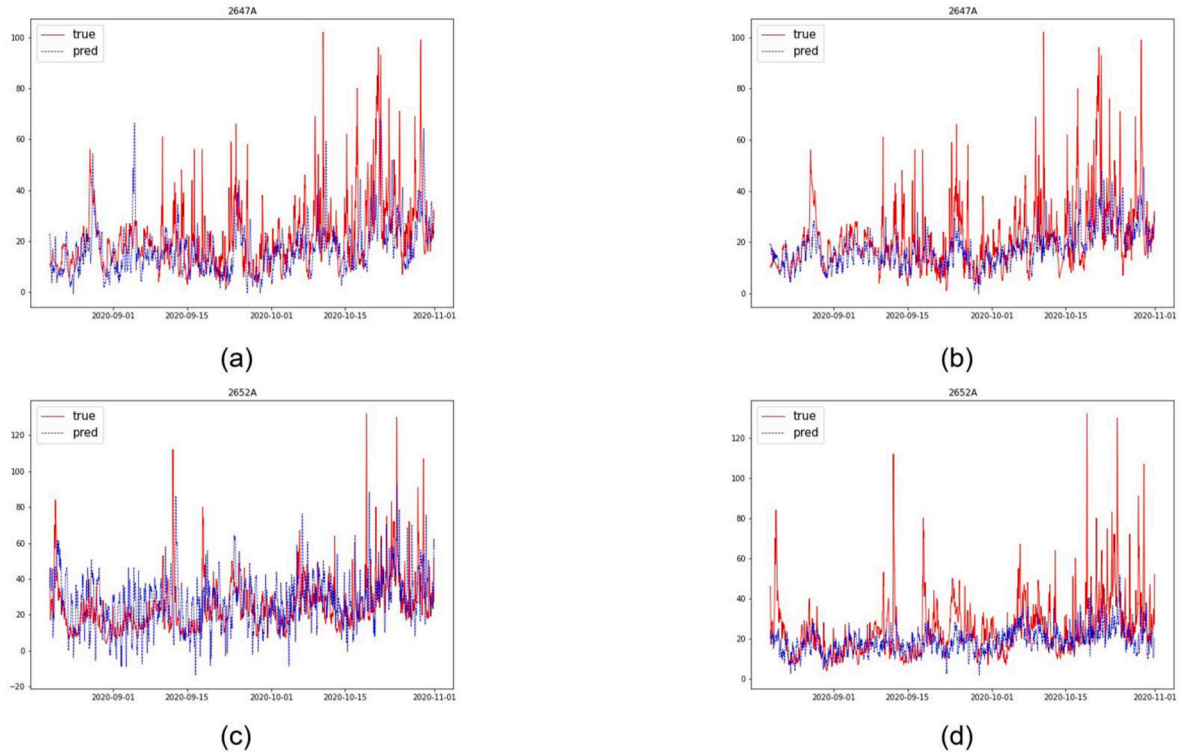


Fig. 10. The 12h prediction results of GC-DCRNN and GLSTM on 2647A and 2652A, where (a) and (c) are the prediction results of GC-DCRNN, and (b) and (d) are the prediction results of GLSTM.

lead to greater deviation between the predicted value and the true mean in formula (20), which will make GC-DCRNN perform better on the metric IA.

4. Conclusion and future work

This paper proposes a Graph-based LSTM (GLSTM) model and predicts hourly PM2.5 in Gansu Province, China. With the combination of parameterized adjacency matrix and LSTM, we introduce spatiotemporal information into the PM2.5 prediction task. By visualizing the

parameterized adjacency matrix obtained by the model on the training set, we can see that the model can introduce spatial information well, and the station has the highest degree of correlation with itself and the surrounding stations. More importantly, the GLSTM model can realize synchronous calculation for all stations, reducing the workload of modeling each station separately. Comparing the PM2.5 heat map predicted by the GLSTM model with the real situation, we can see that the proposed model can predict the overall PM2.5 change in Gansu Province over time.

In the process of constructing the graph, we used the data of the meteorological station closest to the air quality station as the meteorological features of the station. In fact, there is still a certain distance between the meteorological monitoring station and the air quality monitoring station, which means that the original graph is not a homogeneous graph but a heterogeneous graph. Therefore, in future research, we will consider introducing the method of heterogeneous graph neural network into PM2.5 prediction.

Credit author statements

The manuscript does not contain experiments using animals. The manuscript does not contain human studies. All authors vouch that the

work has not been published elsewhere, completely, in part, or any other form, and that the manuscript has not been submitted to another journal. All other authors have read the manuscript and have agreed to submit it in its current form for consideration for publication in the Journal. Meanwhile, we declare that we have no conflict of interest and the data and experiments are real. We further confirm that the order of authors listed in the manuscript has been approved by all of us. We look forward to your positive response.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

The comparison results of the GLSTM model's PM2.5 one-step prediction and the real value at all air quality monitoring stations are shown in Fig. 11(a)–(d), and the comparison results with each model are shown in Table 6 (a)–(d).



Fig. 11 (a). GLSTM model predicts results in one step.

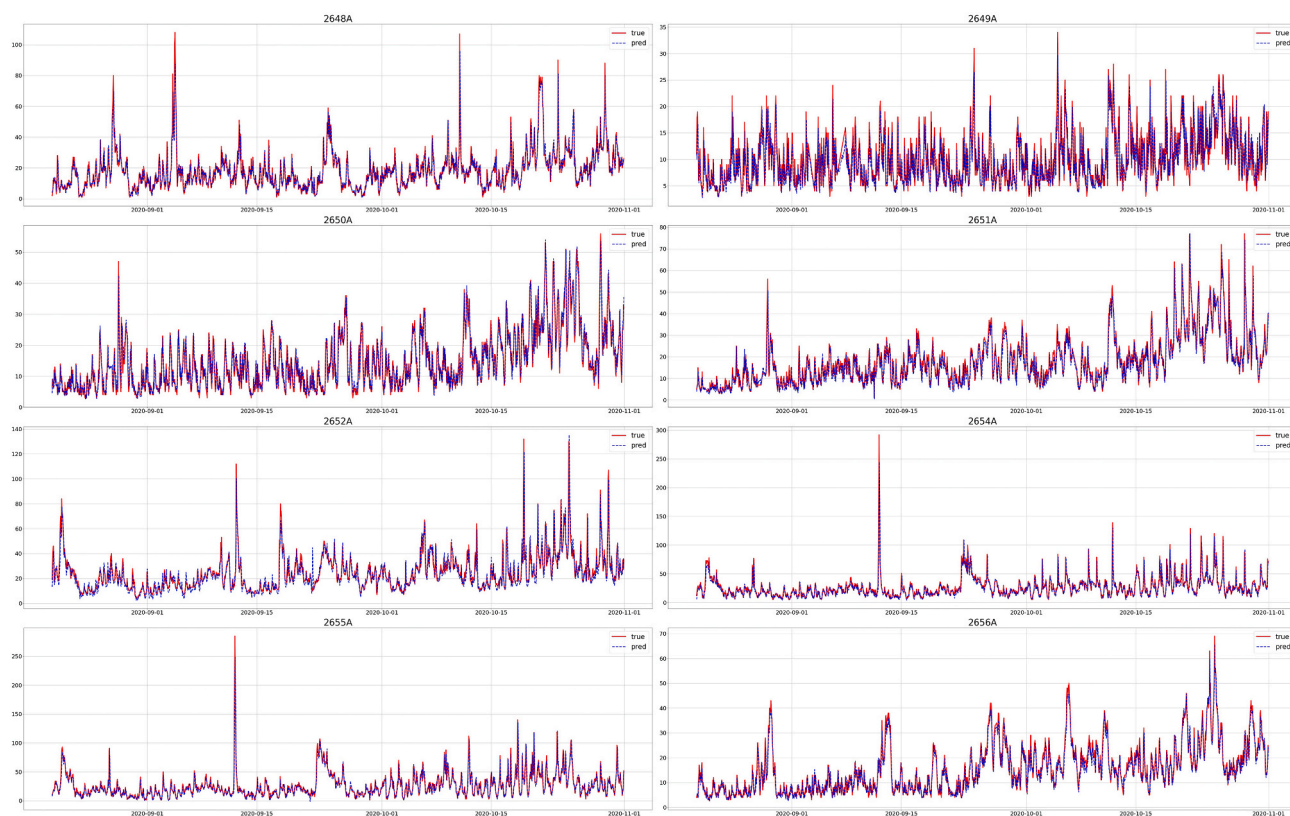


Fig. 11 (b). GLSTM model predicts results in one step.

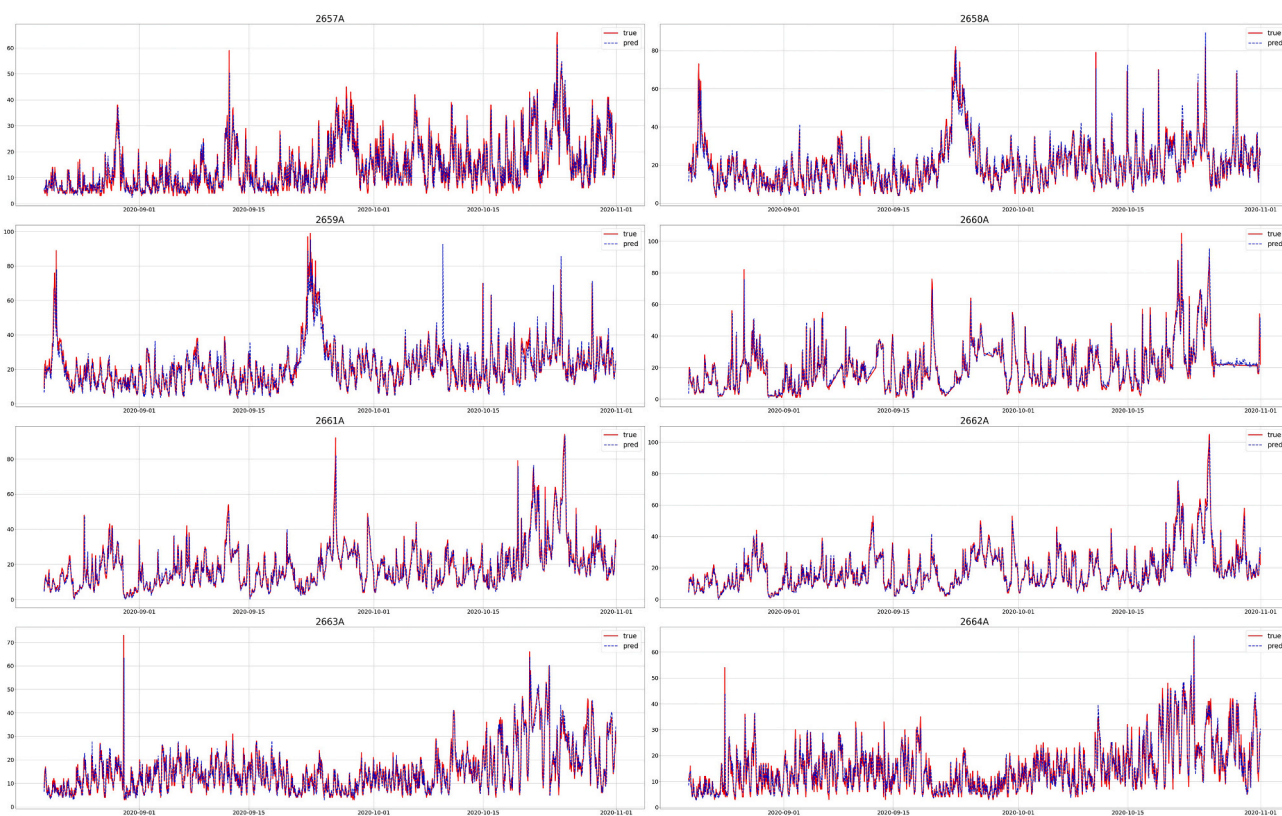


Fig. 11 (c). GLSTM model predicts results in one step.

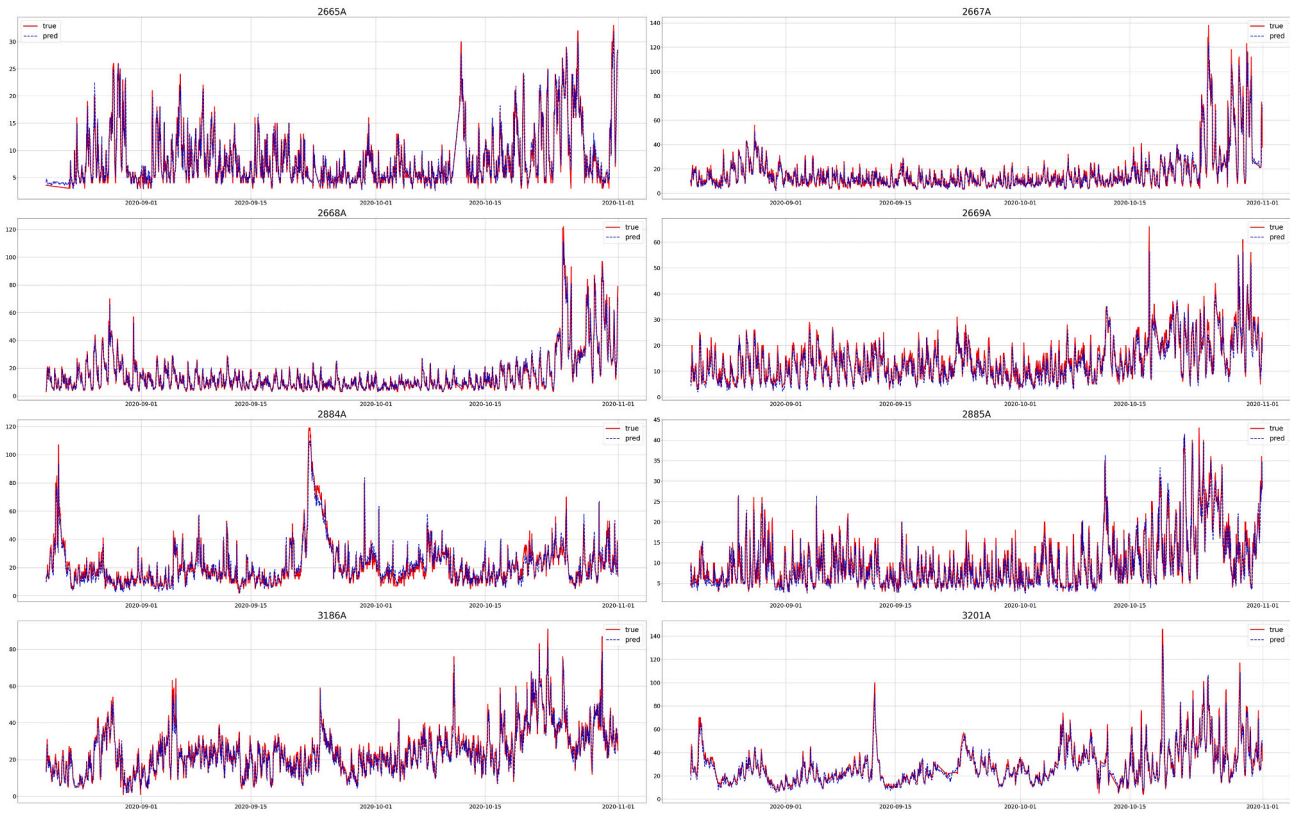


Fig. 11 (d). GLSTM model predicts results in one step

Table 6

(a). PM2.5 one-step prediction model comparison results.

Model	Metric	1476A	1477A	1478A	1479A	1943A	1944A	1945A	2647A
XGBoost	RMSE	6.818	4.164	5.968	7.051	5.994	7.826	7.535	6.224
	MAE	4.852	3.102	4.572	5.598	3.429	4.575	4.489	4.248
	MAPE	0.194	0.206	0.306	0.392	0.359	0.464	0.275	0.244
	IA	0.940	0.917	0.899	0.896	0.939	0.923	0.933	0.939
LSTM	RMSE	6.073 (0.053)	3.618(0.012)	5.111 (0.026)	5.839 (0.017)	5.700(0.256)	5.935 (0.341)	7.141 (0.375)	5.422(0.019)
	MAE	4.391 (0.045)	2.757 (0.007)	3.700 (0.018)	4.307 (0.017)	3.256 (0.199)	3.750 (0.255)	4.650 (0.242)	3.102(0.025)
	MAPE	0.159 (0.001)	0.177 (0.001)	0.198 (0.001)	0.238 (0.004)	0.279 (0.012)	0.375 (0.047)	0.260 (0.014)	0.148(0.002)
	IA	0.950 (0.002)	0.937 (0.001)	0.929 (0.001)	0.929 (0.001)	0.939(0.007)	0.946 (0.007)	0.924 (0.013)	0.955(0.001)
GC-LSTM	RMSE	6.199 (0.199)	3.807 (0.122)	5.008 (0.057)	5.886 (0.066)	6.135 (0.233)	6.278 (0.302)	7.207 (0.850)	6.248 (0.299)
	MAE	4.520 (0.216)	2.878 (0.100)	3.621 (0.029)	4.255 (0.032)	3.464 (0.267)	3.873 (0.448)	5.168 (0.876)	3.939 (0.252)
	MAPE	0.163 (0.006)	0.176 (0.003)	0.196 (0.004)	0.222 (0.004)	0.320 (0.052)	0.405 (0.103)	0.330 (0.073)	0.192 (0.010)
	IA	0.950 (0.003)	0.931 (0.004)	0.935 (0.002)	0.931 (0.002)	0.925 (0.007)	0.936 (0.007)	0.934 (0.014)	0.938 (0.007)
GC-DCRNN	RMSE	5.973(0.103)	3.650 (0.043)	4.978 (0.080)	5.941 (0.095)	6.015 (0.327)	6.070 (0.352)	8.274 (1.001)	11.05 (0.211)
	MAE	4.282(0.100)	2.773 (0.033)	3.574 (0.053)	4.270 (0.053)	3.500 (0.361)	3.796 (0.397)	6.049 (1.049)	7.418 (0.227)
	MAPE	0.156(0.002)	0.176 (0.001)	0.191(0.002)	0.220(0.002)	0.304 (0.015)	0.364 (0.024)	0.386 (0.077)	0.373 (0.012)
	IA	0.953 (0.002)	0.939(0.002)	0.932 (0.002)	0.923 (0.002)	0.932 (0.007)	0.945 (0.006)	0.921 (0.018)	0.806 (0.006)
I-GLSTM	RMSE	6.020 (0.055)	3.696 (0.027)	5.142 (0.024)	6.013 (0.026)	5.940 (0.069)	5.993 (0.138)	7.749 (0.385)	5.593 (0.051)
	MAE	4.306 (0.054)	2.796 (0.022)	3.723 (0.013)	4.348 (0.011)	3.291 (0.081)	3.678 (0.184)	5.390 (0.429)	3.298 (0.061)
	MAPE	0.158 (0.001)	0.176 (0.000)	0.201 (0.002)	0.229 (0.002)	0.304 (0.024)	0.364 (0.047)	0.343 (0.039)	0.160 (0.003)
	IA	0.953 (0.001)	0.936 (0.001)	0.933 (0.001)	0.930 (0.000)	0.935 (0.002)	0.944 (0.003)	0.929 (0.007)	0.953 (0.001)
D-GLSTM	RMSE	6.043 (0.124)	3.682 (0.075)	4.984 (0.076)	5.809 (0.124)	5.980 (0.234)	5.990 (0.308)	6.465 (0.111)	5.941 (0.610)
	MAE	4.379 (0.162)	2.785 (0.060)	3.617 (0.063)	4.249 (0.084)	3.323 (0.162)	3.552 (0.209)	4.406 (0.156)	3.594 (0.536)
	MAPE	0.161 (0.003)	0.174 (0.002)	0.198 (0.006)	0.230 (0.006)	0.275 (0.009)	0.310(0.013)	0.255(0.008)	0.178 (0.035)
	IA	0.952 (0.002)	0.935 (0.002)	0.936 (0.001)	0.933 (0.001)	0.931 (0.006)	0.944 (0.007)	0.945 (0.003)	0.942 (0.016)
GLSTM	RMSE	6.004 (0.070)	3.642 (0.037)	4.864(0.017)	5.661(0.025)	5.762 (0.076)	5.849(0.038)	6.267(0.124)	5.482 (0.019)
	MAE	4.392 (0.075)	2.755(0.032)	3.528(0.012)	4.176(0.027)	3.155(0.058)	3.451(0.030)	4.233(0.126)	3.206 (0.025)
	MAPE	0.160 (0.002)	0.170(0.001)	0.194 (0.002)	0.229 (0.003)	0.275(0.008)	0.316 (0.015)	0.262 (0.016)	0.155 (0.002)
	IA	0.953(0.001)	0.937 (0.002)	0.938(0.001)	0.936(0.000)	0.936 (0.002)	0.946(0.001)	0.948(0.002)	0.954 (0.000)

(b). PM2.5 one-step prediction model comparison results.

Model	Metric	2648A	2649A	2650A	2651A	2652A	2654A	2655A	2656A
XGBoost	RMSE	6.921	3.422	3.907	4.258	7.828	25.011	7.228	3.201
	MAE	4.676	2.629	3.036	3.113	4.552	8.075	4.819	2.491

(continued on next page)

Table 6 (continued)

(b). PM2.5 one-step prediction model comparison results.									
Model	Metric	2648A	2649A	2650A	2651A	2652A	2654A	2655A	2656A
LSTM	MAPE	0.439	0.289	0.274	0.228	0.203	0.252	0.265	0.215
	IA	0.914	0.826	0.949	0.960	0.923	0.740	0.970	0.972
	RMSE	5.525 (0.063)	3.114(0.022)	3.493 (0.030)	3.975(0.058)	10.84 (1.665)	10.01 (0.255)	7.193(0.222)	2.829 (0.028)
	MAE	3.648 (0.070)	2.332(0.010)	2.659 (0.021)	2.760(0.054)	7.275 (1.393)	5.426 (0.285)	4.548(0.272)	2.153 (0.019)
	MAPE	0.262 (0.021)	0.230 (0.002)	0.209 (0.005)	0.180 (0.001)	0.283 (0.040)	0.198 (0.008)	0.243 (0.038)	0.167 (0.001)
GC-LSTM	IA	0.947 (0.001)	0.871 (0.002)	0.960 (0.001)	0.966(0.001)	0.790 (0.082)	0.916 (0.005)	0.969(0.003)	0.978 (0.000)
	RMSE	4.990(0.099)	3.153 (0.045)	3.555 (0.066)	4.372 (0.206)	7.500 (1.239)	9.995 (0.283)	8.434 (0.380)	3.044 (0.192)
	MAE	3.254(0.117)	2.401 (0.071)	2.704 (0.035)	3.059 (0.171)	5.347 (1.417)	5.523 (0.349)	5.198 (0.497)	2.293 (0.135)
	MAPE	0.247 (0.028)	0.247 (0.021)	0.210 (0.007)	0.183 (0.006)	0.261 (0.088)	0.205 (0.022)	0.279 (0.061)	0.167 (0.006)
	IA	0.956(0.002)	0.866 (0.005)	0.959 (0.002)	0.958 (0.004)	0.930 (0.021)	0.919 (0.006)	0.956 (0.005)	0.974 (0.004)
GC-DCRNN	RMSE	9.088 (0.178)	3.377 (0.056)	3.519 (0.044)	4.152 (0.106)	7.720 (0.386)	10.00 (0.427)	8.172 (0.622)	3.033 (0.170)
	MAE	5.885 (0.184)	2.521 (0.038)	2.675 (0.023)	2.885 (0.094)	5.304 (0.537)	5.710 (0.641)	5.218 (0.724)	2.282 (0.135)
	MAPE	0.462 (0.028)	0.252 (0.002)	0.209 (0.003)	0.180 (0.003)	0.250 (0.034)	0.212 (0.028)	0.266 (0.031)	0.165 (0.007)
	IA	0.859 (0.005)	0.868 (0.003)	0.961 (0.001)	0.963 (0.002)	0.933 (0.006)	0.923 (0.006)	0.960 (0.006)	0.974 (0.003)
	RMSE	5.274 (0.043)	3.214 (0.015)	3.539 (0.017)	4.172 (0.052)	7.220(0.386)	9.785(0.085)	8.291 (0.150)	2.808(0.068)
I-GLSTM	MAE	3.405 (0.028)	2.433 (0.027)	2.695 (0.009)	2.877 (0.041)	4.850 (0.472)	5.410 (0.142)	5.056 (0.233)	2.118(0.044)
	MAPE	0.230 (0.008)	0.248 (0.009)	0.211 (0.004)	0.178(0.001)	0.228 (0.031)	0.205 (0.013)	0.264 (0.030)	0.162 (0.003)
	IA	0.953 (0.001)	0.870 (0.001)	0.960 (0.000)	0.962 (0.001)	0.937 (0.007)	0.928(0.002)	0.960 (0.002)	0.979(0.001)
	RMSE	5.455 (0.394)	3.178 (0.035)	3.469 (0.063)	4.197 (0.099)	6.734 (0.274)	10.14 (0.228)	8.571 (0.567)	2.912 (0.112)
	MAE	3.563 (0.351)	2.370 (0.024)	2.649 (0.051)	2.939 (0.104)	4.279 (0.312)	5.574 (0.233)	5.271 (0.455)	2.193 (0.078)
D-GLSTM	MAPE	0.264 (0.065)	0.232 (0.007)	0.210 (0.005)	0.182 (0.005)	0.180 (0.014)	0.197 (0.008)	0.248 (0.017)	0.161 (0.002)
	IA	0.946 (0.010)	0.868 (0.004)	0.961 (0.001)	0.962 (0.002)	0.941 (0.007)	0.917 (0.007)	0.955 (0.007)	0.976 (0.002)
	RMSE	5.089 (0.034)	3.149 (0.025)	3.405(0.020)	4.109 (0.058)	6.458(0.148)	9.924 (0.133)	8.215 (0.126)	2.849 (0.055)
	MAE	3.289 (0.034)	2.345 (0.010)	2.606(0.017)	2.892 (0.058)	3.864(0.184)	5.285(0.173)	4.922 (0.165)	2.150 (0.036)
	MAPE	0.227(0.005)	0.229(0.002)	0.209(0.003)	0.181 (0.003)	0.161(0.009)	0.188(0.004)	0.238(0.006)	0.160(0.001)
GLSTM	IA	0.955 (0.000)	0.871(0.002)	0.963(0.000)	0.964 (0.001)	0.946(0.003)	0.920 (0.003)	0.960 (0.001)	0.977 (0.001)
(c). PM2.5 one-step prediction model comparison results.									
Model	Metric	2657A	2658A	2659A	2660A	2661A	2662A	2663A	2664A
XGBoost	RMSE	5.684	6.334	6.736	6.246	6.003	4.890	5.310	5.184
	MAE	4.290	4.010	4.168	4.314	4.169	3.514	3.766	3.994
	MAPE	0.420	0.241	0.223	0.455	0.414	0.324	0.355	0.411
	IA	0.902	0.925	0.923	0.941	0.939	0.961	0.915	0.908
	RMSE	4.854 (0.205)	5.068(0.262)	8.074 (0.314)	5.367 (0.020)	4.471 (0.031)	3.764 (0.099)	4.265 (0.026)	4.350 (0.055)
LSTM	MAE	3.464 (0.108)	3.410(0.199)	3.972 (0.389)	3.113 (0.031)	2.701 (0.047)	2.530 (0.107)	2.818 (0.022)	3.062 (0.038)
	MAPE	0.253(0.005)	0.187 (0.015)	0.203 (0.029)	0.194(0.002)	0.166 (0.003)	0.158 (0.007)	0.223 (0.003)	0.234 (0.003)
	IA	0.927 (0.009)	0.945 (0.011)	0.877 (0.020)	0.958 (0.000)	0.967 (0.001)	0.976 (0.001)	0.946 (0.001)	0.938 (0.002)
	RMSE	4.715 (0.049)	5.625 (0.413)	5.966 (0.547)	5.507 (0.073)	4.704 (0.236)	3.899 (0.097)	4.424 (0.103)	4.145 (0.168)
	MAE	3.423 (0.068)	3.967 (0.399)	4.186 (0.519)	3.323 (0.116)	2.957 (0.289)	2.648 (0.100)	2.923 (0.085)	3.111 (0.218)
GC-LSTM	MAPE	0.267 (0.022)	0.221 (0.030)	0.218 (0.033)	0.248 (0.039)	0.187 (0.022)	0.176 (0.020)	0.226 (0.012)	0.265 (0.040)
	IA	0.932 (0.002)	0.935 (0.008)	0.938 (0.010)	0.955 (0.002)	0.963 (0.004)	0.974 (0.002)	0.939 (0.003)	0.943 (0.005)
	RMSE	4.689 (0.048)	5.973 (0.762)	6.614 (0.875)	5.427 (0.117)	4.729 (0.203)	3.891 (0.196)	4.459 (0.167)	4.023(0.068)
	MAE	3.369 (0.013)	4.349 (0.799)	4.768 (1.000)	3.226 (0.212)	2.953 (0.340)	2.684 (0.246)	2.981 (0.164)	2.935(0.040)
	MAPE	0.254 (0.006)	0.240 (0.049)	0.250 (0.062)	0.207 (0.015)	0.196 (0.033)	0.173 (0.023)	0.221 (0.009)	0.229(0.008)
I-GLSTM	IA	0.935 (0.001)	0.931 (0.016)	0.928 (0.018)	0.958 (0.002)	0.965 (0.003)	0.976 (0.002)	0.939 (0.004)	0.949(0.002)
	RMSE	4.848 (0.032)	5.675 (0.185)	6.542 (0.200)	5.422 (0.016)	4.539 (0.060)	3.749 (0.023)	4.455 (0.024)	4.303 (0.038)
	MAE	3.490 (0.020)	3.910 (0.194)	4.318 (0.219)	3.097 (0.035)	2.725 (0.094)	2.499 (0.029)	2.881 (0.016)	3.095 (0.049)
	MAPE	0.262 (0.008)	0.221 (0.018)	0.225 (0.017)	0.207 (0.014)	0.168 (0.006)	0.161 (0.008)	0.225 (0.004)	0.237 (0.013)
	IA	0.932 (0.001)	0.938 (0.004)	0.930 (0.004)	0.958 (0.000)	0.966 (0.001)	0.977 (0.000)	0.943 (0.001)	0.941 (0.002)
D-GLSTM	RMSE	4.705 (0.100)	5.245 (0.091)	5.715 (0.124)	5.363 (0.125)	4.526 (0.088)	3.785 (0.112)	4.352 (0.087)	4.147 (0.135)
	MAE	3.405 (0.045)	3.497 (0.080)	3.768 (0.131)	3.091 (0.103)	2.796 (0.104)	2.524 (0.103)	2.833 (0.043)	3.009 (0.095)
	MAPE	0.258 (0.007)	0.186(0.007)	0.185 (0.009)	0.203 (0.007)	0.169 (0.006)	0.155 (0.006)	0.218 (0.005)	0.234 (0.011)
	IA	0.934 (0.002)	0.944 (0.001)	0.941 (0.004)	0.958 (0.002)	0.966 (0.001)	0.976 (0.002)	0.944 (0.002)	0.946 (0.003)
	RMSE	4.580(0.016)	5.107 (0.076)	5.482(0.082)	5.260(0.039)	4.375(0.017)	3.676(0.025)	4.226(0.015)	4.064 (0.037)
GLSTM	MAE	3.383 (0.017)	3.462 (0.094)	3.536(0.067)	3.091(0.064)	2.670(0.021)	2.445(0.022)	2.761(0.013)	3.005 (0.044)
	MAPE	0.268 (0.005)	0.199 (0.011)	0.184(0.007)	0.216 (0.013)	0.165(0.002)	0.152(0.003)	0.217(0.003)	0.241 (0.008)
	IA	0.937(0.001)	0.946(0.002)	0.945(0.002)	0.960(0.001)	0.968(0.000)	0.978(0.000)	0.947(0.001)	0.948 (0.001)
(d). PM2.5 one-step prediction model comparison results.									
Model	Metric	2665A	2666A	2667A	2668A	2669A	2884A	2885A	3201A
XGBoost	RMSE	3.158	7.585	5.432	3.789	8.924	3.927	6.266	7.653
	MAE	2.514	4.943	3.749	2.947	4.986	2.872	4.825	4.818
	MAPE	0.405	0.409	0.365	0.266	0.283	0.344	0.290	0.185
	IA	0.900	0.934	0.963	0.937	0.919	0.899	0.926	0.943
	RMSE	2.414 (0.015)	6.285(0.036)	4.874(0.103)	3.570(0.029)	8.597 (1.738)	3.184(0.021)	5.514(0.010)	6.866 (0.696)
LSTM	MAE	1.678 (0.006)	3.779 (0.027)	3.163 (0.061)	2.607(0.017)	5.794 (1.036)	2.301 (0.020)	4.056(0.011)	4.276 (0.481)
	MAPE	0.203 (0.002)	0.265 (0.006)	0.255 (0.006)	0.208 (0.003)	0.303 (0.054)	0.239 (0.005)	0.210 (0.002)	0.141 (0.015)
	IA	0.941 (0.002)	0.957 (0.001)	0.970(0.002)	0.942 (0.002)	0.879 (0.077)	0.935 (0.001)	0.942 (0.000)	0.945 (0.016)
	RMSE	2.435 (0.017)	6.466 (0.069)	5.371 (0.153)	3.853 (0.201)	7.182 (0.636)	3.237 (0.028)	5.875 (0.166)	6.448 (0.565)
	MAE	1.675(0.022)	3.833 (0.083)	3.240 (0.090)	2.776 (0.194)	5.173 (0.598)	2.305 (0.049)	4.285 (0.147)	4.369 (0.585)
GC-LSTM	MAPE	0.200(0.009)	0.250 (0.017)	0.245 (0.018)	0.220 (0.024)	0.284 (0.044)	0.235 (0.015)	0.195 (0.004)	0.157 (0.026)
	IA	0.942 (0.001)	0.961(0.001)	0.962 (0.003)	0.937 (0.007)	0.939 (0.010)	0.935 (0.001)	0.938 (0.003)	0.954 (0.008)
	RMSE	2.516 (0.020)	7.042 (0.047)	6.521 (0.114)	4.221 (0.264)	7.967 (1.195)	3.427 (0.028)	5.867 (0.140)	6.781 (0.941)

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Table 6 (continued)

(d). PM2.5 one-step prediction model comparison results.									
Model	Metric	2665A	2666A	2667A	2668A	2669A	2884A	2885A	3201A
I-GLSTM	MAE	1.727 (0.015)	4.538 (0.093)	3.915 (0.045)	3.049 (0.248)	5.977 (1.208)	2.460 (0.030)	4.289 (0.127)	4.852 (1.073)
	MAPE	0.205 (0.002)	0.308 (0.007)	0.288 (0.010)	0.233 (0.018)	0.335 (0.072)	0.255 (0.007)	0.193 (0.003)	0.180 (0.045)
	IA	0.942 (0.001)	0.954 (0.001)	0.936 (0.002)	0.928 (0.009)	0.932 (0.019)	0.932 (0.001)	0.935 (0.003)	0.952 (0.013)
	RMSE	2.450 (0.008)	6.723 (0.029)	5.172 (0.021)	3.746 (0.047)	6.974 (0.276)	3.268 (0.013)	5.713 (0.040)	5.850 (0.199)
	MAE	1.690 (0.012)	3.946 (0.026)	3.115 (0.014)	2.677 (0.054)	4.949 (0.281)	2.339 (0.026)	4.117 (0.034)	3.914 (0.212)
	MAPE	0.202 (0.005)	0.256 (0.009)	0.226(0.006)	0.207 (0.010)	0.280 (0.026)	0.241 (0.007)	0.195 (0.000)	0.145 (0.011)
D-GLSTM	IA	0.943 (0.000)	0.956 (0.000)	0.967 (0.000)	0.942(0.002)	0.947 (0.004)	0.935 (0.000)	0.942 (0.001)	0.966 (0.003)
	RMSE	2.422 (0.029)	6.738 (0.208)	5.234 (0.140)	4.102 (0.231)	6.829 (0.396)	3.226 (0.016)	5.664 (0.079)	6.044 (0.369)
	MAE	1.684 (0.020)	3.995 (0.221)	3.185 (0.105)	2.914 (0.173)	4.687 (0.289)	2.280 (0.015)	4.114 (0.081)	3.958 (0.359)
	MAPE	0.205 (0.006)	0.252 (0.016)	0.232 (0.009)	0.215 (0.008)	0.242(0.008)	0.227 (0.004)	0.195 (0.004)	0.137 (0.013)
GLSTM	IA	0.943 (0.001)	0.956 (0.001)	0.965 (0.002)	0.929 (0.009)	0.945 (0.008)	0.935 (0.001)	0.943 (0.002)	0.962 (0.005)
	RMSE	2.393(0.007)	6.463 (0.030)	5.096 (0.034)	3.928 (0.100)	6.731(0.110)	3.189 (0.020)	5.590 (0.054)	5.733(0.163)
	MAE	1.677 (0.009)	3.776(0.013)	3.113(0.020)	2.776 (0.079)	4.679(0.143)	2.248(0.011)	4.077 (0.048)	3.617(0.188)
	MAPE	0.207 (0.003)	0.242(0.003)	0.234 (0.004)	0.204(0.003)	0.266 (0.019)	0.224(0.001)	0.191(0.001)	0.126(0.006)
	IA	0.946(0.000)	0.958 (0.001)	0.967 (0.001)	0.933 (0.004)	0.948(0.002)	0.937(0.001)	0.945(0.001)	0.966(0.002)

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